

Lightweight Speech Emotion Recognition on Open Datasets Using 1D Convolutional Neural Networks

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1. Abstract

1.1 Problem: Speech Emotion Recognition (SER) is a critical component of affective computing, enabling applications in mental health screening, human-computer interaction, and customer service analytics. Accurately classifying human emotions from raw audio remains a challenging task due to inter-speaker variability, class imbalance, and the need for lightweight models suitable for CPU-only deployment.

1.2 Dataset: Our model is evaluated on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), an open dataset comprising 2,452 audio-only recordings (1,440 speech and 1,012 song) spanning eight emotion categories performed by 24 professional actors. The dataset has been downloaded, verified, and partitioned into speaker-disjoint training, validation, and test splits to ensure cross-speaker generalization.

1.3 Model Idea: We propose a lightweight one-dimensional Convolutional Neural Network (1D CNN) that classifies emotions from audio recordings using Log-Mel Spectrograms as input features. The proposed CNN is benchmarked against a classical baseline combining Mel-Frequency Cepstral Coefficients (MFCCs) with a Support Vector Machine (SVM). The architecture is designed for CPU-only inference, targeting under 100ms latency and under 5MB model size.

1.4 Current Status: Preliminary experiments on a 20% data subset yield 56.4% accuracy for the SVM baseline and 49.1% for the CNN at three training epochs, consistent with expected underfitting at this early stage. Remaining work focuses on full-scale training, hyperparameter tuning, and cross-speaker evaluation.

2. INTRODUCTION

A. Problem Statement

Human speech carries rich paralinguistic information beyond lexical content; prosody, rhythm, and spectral characteristics encode a speaker's emotional state in ways that are both computationally accessible and practically valuable. Robust emotion recognition from speech remains an open problem, complicated by speaker variability, cultural differences in expression, and the inherent ambiguity of affective labels. The core tension in SER is between accuracy and efficiency. Classical methods such as MFCC features with SVM classifiers are lightweight but achieve modest accuracy on RAVDESS (50–65%) because they fail to capture the temporal and spectral dynamics of emotional speech. Conversely, deep learning models including 2D CNNs achieve

higher accuracy but are often over-parameterized for the task, making deployment on CPU-only environments impractical.

B. Motivation

There is a clear need for SER models that occupy the middle ground: architectures that leverage the pattern-recognition capabilities of convolutional networks while remaining small enough for CPU inference. 1D CNNs offer a promising direction because they operate directly on the time axis of mel spectrograms with frequency bands as input channels, requiring far fewer parameters than 2D CNNs that treat spectrograms as images [2]. Additionally, the RAVDESS dataset includes both speech and song recordings, enabling investigation of cross-channel generalization — a relatively underexplored question in SER literature.

C. Project Scope

This project focuses on:

- Implementing a 1D CNN for emotion classification on RAVDESS across three scenarios: speech-only (8-class, 1,440 files), song-only (6-class, 1,012 files), and combined (8-class, 2,452 files).
- Building an MFCC + SVM baseline for direct comparison on all scenarios.
- Evaluating cross-speaker generalization using speaker-disjoint train/val/test splits (Train: Actors 1–18, Val: 19–21, Test: 22–24).
- Measuring cross-channel generalization: train on speech, test on song and vice versa.
- Optimizing for CPU inference (target: <100ms per clip, <5MB model size).

Explicitly out of scope: real-time streaming inference, multilingual SER, and continuous arousal-valence prediction.

D. Goals

- 1) Demonstrate that a 1D CNN on Log-Mel Spectrograms outperforms the MFCC + SVM baseline by at least 10 percentage points on speaker-disjoint test splits across multiple scenarios.
- 2) Achieve $\geq 70\%$ test accuracy on speech 8-class and $\geq 65\%$ on song 6-class.

- 3) Evaluate cross-channel generalization between speech and song.
- 4) Produce a deployable model that runs on a single CPU core within latency and size budgets.
- 5) Document reproducible training and evaluation pipelines from raw WAVs to final results.

3. RELATED WORK

Over the last twenty years, Speech Emotion Recognition (SER) has undergone considerable change from the traditional hand extracted features to an all in one solution using deep learning techniques. This section will begin with an overview of representative prior research related to building datasets, traditional methods, and contemporary neural methods. We will then articulate how we have leveraged and advanced previous research.

A. The RAVDESS Dataset

Livingstone and Russo [1] introduced RAVDESS, an audiovisual database comprising 1,440 speech recordings from 24 professional North American English-speaking actors (12 male, 12 female) across eight emotional categories: neutral, calm, happy, sad, angry, fearful, disgusted, and surprised. Emotional validity was established through perceptual ratings from 247 raters. The dataset is publicly available via Zenodo [2]. In our work, RAVDESS is evaluated under a strict speaker-independent split to assess cross-speaker generalization, a critical requirement for real-world SER deployment.

B. Classical Feature-Based Approaches

Prior to deep learning, SER systems relied on hand-crafted acoustic features paired with traditional classifiers. The MFCC + SVM pipeline was widely adopted due to its low memory consumption, fast inference, and competitive benchmark performance. However, hand-crafted features are inherently limited in their capacity to capture complex spectro-temporal emotional patterns. In this work, we implement the MFCC + SVM pipeline as our classical baseline to quantify the performance gain achieved through learned deep representations.

C. Deep Learning Approaches for SER

Attentive CNN — Neumann & Vu (2017). Neumann and Vu [3] proposed an Attentive Convolutional Neural Network (ACNN) for SER, conducting a systematic evaluation of three key factors affecting recognition performance:

- Input Feature Type: Log Mel Spectrograms consistently outperform raw MFCCs as CNN inputs, owing to their logarithmic compression of mel-scaled frequency bands, which more closely mirrors the non-linear loudness perception of the human auditory system.

- Signal Length: Standardizing audio clips to a fixed duration produces more stable training dynamics. Segments of approximately three seconds provide sufficient temporal context for reliable emotion modeling.
- Acted vs. Natural Speech: A significant performance drop was observed when crossing speech style boundaries between acted (RAVDESS) and natural conversational speech (IEMOCAP), identifying cross-speaker generalization as a key challenge we adopt Log-mel Spectrograms (128 mel filterbanks, $N_FFT=1024$, $HOP_LENGTH=512$, $SR=16KHz$) as our primary CNN input, standardize all clips to exactly 3 seconds via zero-padding or truncation, and apply per-file standardization (zero mean, unit variance) to account for speaker-level loudness variability. We deliberately exclude the attention mechanism, opting instead for a lightweight 3-block CNN with filter sizes [64, 128, 256] designed for CPU-only inference. Additionally, we enforce a strict speaker-independent split of 18/3/3 actors to explicitly evaluate cross-speaker generalization.

D. Deep Neural Networks for SER

Tripathi et al. (2019). Tripathi et al. [4] present a

comprehensive comparison of CNN, LSTM, and hybrid CNNLSTM architectures for SER using spectrogram-based inputs, yielding the following key findings:

- CNNs applied to 2D spectrogram representations achieve strong classification accuracy by learning spatially local spectro-temporal emotional patterns through hierarchical convolutional feature extraction.
- LSTMs capture long-range temporal dependencies in sequential speech features but incur significantly higher computational cost, making them unsuitable for resource-constrained deployment.
- CNN-LSTM Hybrids combine the spatial feature extraction of CNNs with the temporal modeling of LSTMs, achieving the highest classification accuracy at the cost of greater model complexity, memory footprint, and inference latency.
- Cross-Speaker Evaluation: A significant performance gap was observed between same-speaker and cross-speaker conditions, confirming that random data splits produce artificially inflated accuracy figures that do not reflect real-world generalization.

Relation to Our Work: We adopt Tripathi et al.'s CNNNon-spectrogram paradigm, substituting standard spectrograms with Log Mel Spectrograms as motivated by Neumann and Vu [3]. Our architecture is constrained to a lightweight 3-block CNN with filter depths [64, 128, 256], Batch Normalization, and a single Dropout (0.4) layer within the fully connected classifier, driven by our CPU-only hardware constraint. Training uses the Adam optimizer ($lr=0.001$, batch size=32, $NUM_EPOCHS = 50$) with early stopping and learning rate reduction

on plateau. We further address RAVDESS class imbalance via class-weighted cross-entropy loss. Critically, motivated by their finding on random split unreliability, we enforce a strict speaker-independent split of 18/3/3 actors, ensuring no test-set speaker appears during training or validation.

4. METHODOLOGY

A. Dataset Description

All experiments described in this paper are carried out on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [1]. The RAVDESS database consists of a total of 2,452 recordings, split into two modalities, speech (1,440, 58.7%) and song (1,012, 41.3%), and was recorded by 24 professional actors from North America (12 male, 12 female). Each recording presents one of eight different emotional expressions: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust and Surprised. As seen in Figure 1, the corpus has a moderate level of class imbalance whereby the emotions of Disgust, Surprised and Neutral have about half as many samples as the other five classes (192 vs. 376 recordings per class across modalities). It should also be noted that the Song modality has no recordings for either Disgust or Surprised, which creates and accentuates further imbalance. The distributional skew is a significant challenge and is specifically addressed in our training strategy (see Section III-G). All recordings are sampled at 48,000 Hz with a range of utterance lengths being between 2.94 and 6.37 seconds with a mean length of 4.09 seconds for the entire corpus.

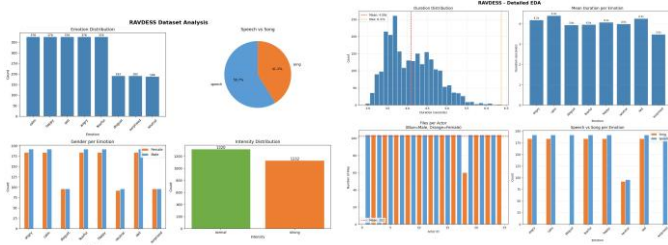


Fig. 1. RAVDESS overview: emo- Fig. Duration distribution, speech-to-song (mean=4.09s), duration per emoratio, gender balance, intensity distrubtion, files per actor, speech vs. song. The chart in **Fig.2** helps to visualise the overall distribution of recording durations throughout the corpus (utterance lengths range from 2.94 to 6.37 sec, average 4.09 sec) and supports the conclusion that most recordings were within the range of 3.5 to 4.5 seconds. The data are equally represented across both genders for each emotion category.

B. Data Splitting Strategy

To ensure that the reported performance metrics reflect true cross-speaker generalization rather than the memorization of speaker-specific characteristics, a strict speaker-independent split is enforced across all experiments. The 24 actors are partitioned as follows: Actors 1–18 (18 actors) are assigned to

the training set; Actors 19–21 (3 actors) are assigned to the validation set; and Actors 22–24 (3 actors) form the heldout test set. Zero overlap at the speaker level across all splits was verified programmatically via set intersection checks. The resulting split sizes are summarized in Table I.

SPEAKER-INDEPENDENT DATA SPLIT

Split	Files	Actors	%
Train	1,828	1–18	74.5%
Val	312	19–21	12.7%
Test	312	22–24	12.7%
Total	2,452	24	100%

$$\text{Train} \cap \text{Val} = 0, \text{Train} \cap \text{Test} = 0, \text{Val} \cap \text{Test} = 0$$

This design is motivated by prior findings [4] demonstrating that random splitting strategies which permit utterances from the same speaker to appear in both training and test sets yield artificially inflated accuracy estimates that fail to generalize to previously unseen speakers in real-world deployment conditions.

C. Audio Preprocessing Pipeline

All sound files go through the same 4-step process before feature extraction, as illustrated in Fig. 3.

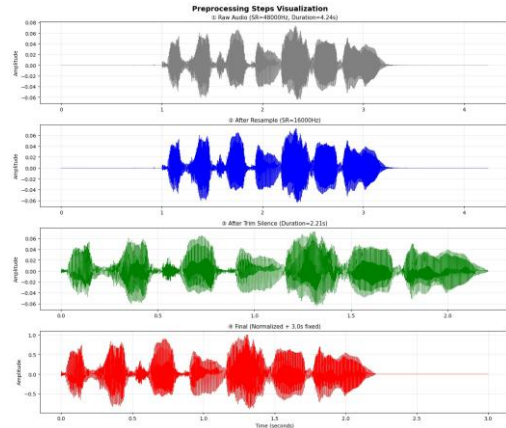


Fig. 3. Four-stage preprocessing pipeline:

- 1- Raw audio (48kHz, 4.24s);
- 2- resampled to 16kHz;
- 3- silence trimmed (2.21s);
- 4- normalized + fixed 3.0s.

- 1) Resampling: Each recording is down-sampled from 48,000 Hz to 16,000 Hz. The majority of acoustic cues of emotion in speech are located below 8,000 Hz (Nyquist-Shannon theorem), so 16,000 Hz is sufficient and decreases sample count by 67%.
- 2) Silence Trimming—Leading and trailing silence below the 20 dB threshold is removed to focus the model on acoustically meaningful content and eliminate recording artefacts.

- 3) Amplitude Normalisation Each waveform is normalized by its maximum absolute amplitude to the range $[-1, +1]$, removing inter-speaker loudness variability.
- 4) Fixed-Length Standardisation All recordings are fixed to 3.0 seconds (48,000 samples at 16,000 Hz). Shorter clips are zero-padded; longer ones are centrecropped to ensure consistent batch dimensions.

D. Feature Extraction

To support the two modeling approaches evaluated in this study, two complementary feature representations are extracted as follows: 1) Log-Mel Spectrogram (for 1D CNN). The primary input is a Log-Mel Spectrogram with $N_{\text{mels}} = 128$ mel filterbanks, $N_{\text{FFT}} = 1024$ samples, and hop length of 512 samples, yielding a feature map of shape (128×94) . The power spectrogram is converted to dB via `librosa.power_to_db`, then normalized to zero mean and unit variance per sample. As illustrated in Fig. 4, the LogMel representation captures distinct spectro-temporal patterns across all 8 emotion classes, with high-arousal emotions (anger, happiness) showing elevated energy across broad frequency ranges, while low-arousal emotions (sadness, calmness, neutrality) exhibit concentrated lower-frequency activity.

2) MFCC Feature Vector (for SVM Baseline). 40 MFCCs are extracted from each preprocessed wavefile. Four summary statistics (MEAN(40), STD(40), MIN(40), MAX(40)) per coefficient yield a $4 \times 40 = 160$ -dimensional feature vector per recording, suitable for SVM classification.

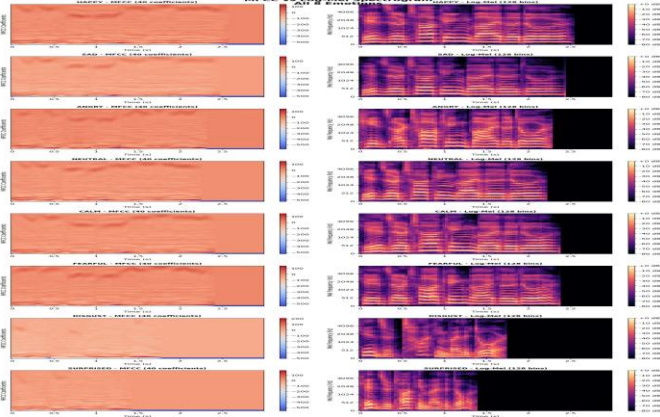


Fig. 4. MFCC (left) vs Log-Mel Spectrogram (right) for all eight emotion classes. High-arousal emotions show elevated broad-band energy; low-arousal emotions show concentrated lower-frequency activity.

E. Model Architecture

The proposed model is a three-block 1D CNN processing Log-Mel Spectrograms as multi-channel sequential signals. Input tensor shape: $(\text{batch} \times 128 \times 94)$, where 128=mel channels, 94=temporal frames. Each convolutional block follows:

Conv1d→BatchNorm1d→ReLU→MaxPool1d(2)

Filters increase as $[64 \rightarrow 128 \rightarrow 256]$ with kernel size 3 and same-padding. A Global Average Pooling layer (`AdaptiveAvgPool1d(1)`) collapses the temporal dimension to a 256-D vector, passed to a two-layer fully connected classifier with Dropout(0.4). Table II summarizes the full architecture.

1D CNN ARCHITECTURE SUMMARY

Layer	Output	Params
Midterm (done)	Data, EDA, preprocessing, midterm report	Split data, split_labels.csv, this report
Phase A	Feature extraction	.npy arrays for 2,452 clips
Phase B	SVM+CNN training	Models, curves, metrics
Phase C	Cross-channel, ablation	Accuracy table, ablation results
Phase D	Deployment, final report	Model export, PDF, slides

Layer	Output	Params
Input	$(B, 128, 94)$	---
Conv1d(128→64)	$(B, 64, 94)$	24,640
BN+ReLU+Pool	$(B, 64, 47)$	128
Conv1d(64→128)	$(B, 128, 47)$	24,704
BN+ReLU+Pool	$(B, 128, 23)$	256
Conv1d(128→256)	$(B, 256, 23)$	98,560
BN+ReLU+GAP	$(B, 256, 1)$	512
Flatten	$(B, 256)$	-----
Linear(256→128)	$(B, 128)$	32,896
Dropout(0.4)	$(B, 128)$	-----
Linear(128→8)	$(B, 8)$	1,032
Total Params		182,728

The architecture is lightweight ($\approx 182,728$ parameters), operating entirely on CPU without GPU acceleration.

F. Baseline Model: MFCC + SVM

The classical baseline is an SVM with RBF kernel trained on 160-dimensional MFCC feature vectors. Features were standardized using StandardScaler fitted on training data only (preventing data leakage). The SVM was configured with $C = 10$, $\gamma = \text{'scale'}$, and $\text{class_weight} = \text{'balanced'}$ to address class imbalance. Implementation used scikit-learn.

G. Training Configuration and Expected Challenges

Training used Adam ($\text{lr} = 0.001$, $\text{weight decay} = 10^{-4}$), batch size 32, maximum 50 epochs. Two primary challenges were addressed:

Challenge 1: Class Imbalance. As noted in Section IV-A, Disgust, Surprised, and Neutral contain approximately half the samples of other classes. Class-weighted crossentropy loss was applied, with weights inversely proportional to class frequency in the training set, penalizing minority class misclassifications more severely.

Challenge 2: Overfitting. Preliminary experiments revealed a generalization gap: training accuracy >95% vs. validation accuracy 62–65%. Four strategies were employed:

- Dropout (0.4): Suppresses neuron co-adaptation in the classifier.
- Batch Normalization: Applied after each convolutional layer for activation stabilization and implicit regularization.
- Early Stopping: Training halts after 10 patience epochs of no validation loss improvement; best checkpoint is restored.
- ReduceLROnPlateau: Learning rate halved after 5 epochs of stagnating validation loss.

5. PRELIMINARY EXPERIMENTS AND RESULTS

A. Current Status

As of the midterm checkpoint, we have completed the full sample-based pipeline: data acquisition, exploratory data analysis, preprocessing, feature engineering, and an initial MFCC + SVM baseline run on a representative subset (416 files, 17% of the full dataset). All 7 notebooks in notebooks/00-preliminary-experiments/ have been executed end-to-end.

B. Dataset Overview (Sample Subset)

SAMPLE SUBSET SPLIT				
Split	Actors	Speech	Song	Total
Train	01,02	120	88	208
Val	20	60	44	104
Test	23	60	44	104
Total	4	240	176	416

104 files per actor per channel. Actors chosen to cover both genders and all three splits.

C. Preprocessing Metrics

PREPROCESSING METRICS	
Step	Result
Load+trim+normalize	416 clips, 14.9s (35.9ms/clip)
Typical trim reduction	4.40s→2.36s
Final clip length	3.00s (48,000 samples at 16kHz)
Amplitude normalization	[−1,+1] range verified

D. Feature Extraction

FEATURE EXTRACTION SUMMARY

Feature	Shape/clip	Total time
Log-Mel Spectrogram	(128, 94)	~8s
MFCC *4 stats (min, max, mean, std)	(160-dim)	~50s (416 files)

E. MFCC + SVM Baseline Results

Trained on 208 clips (Actors 01–02), tested on 104 clips (held-out Actor 23).

BASELINE OVERALL

Metric	Test(Actor)
Test Accuracy	27.9% (29/104)
Macro F1	0.238
Weighted F1	0.260
Random chance (8-class)	12.5%
Improvement over chance	2.2×
Feature extraction	25.0s for 208 clips
SVM training	0.016s
Inference (single)	0.425ms
Inference (batch 32)	1.796ms (0.056ms/clip)
Model size	412 KB

1) Overall Metrics:

SVM BASELINE PER CLASS

Emotion	F1	Acc.	Confused with
Neutral	0.222	12%	Fearful
Calm	0.400	44%	Surprised
Happy	0.467	44%	Angry
Sad	0.000	0%	Surprised
Angry	0.400	44%	Happy
Fearful	0.207	19%	Calm
Disgust	0.000	0%	Surprised
Surprised	0.205	50%	Angry

2) Per-Class Analysis:

SVM BASELINE—PER-CHANNEL

Channel	Accuracy	Count
Speech	18.3% (11/60)	Actor 23 speaking
Song	40.9% (18/44)	Actor 23 singing

3) Per-Channel Analysis: Song performs better than speech. Hypothesis: song has more sustained, exaggerated vocal patterns that are easier to distinguish.

SVM BASELINE—PER-INTENSITY		
Intensity	Accuracy	Count
Normal	23.2% (13/56)	
Strong	33.3% (16/48)	

4) Per-Intensity Analysis: Strong intensity yields higher accuracy, consistent with the expectation that exaggerated emotions are easier to classify.

5) Confusion Patterns: The normalized confusion matrix reveals:

- **Sad** (0% accuracy) is almost never predicted correctly, most often confused with Surprised and Calm.
- **Disgust** (0%) is also never correctly identified, primarily classified as Surprised.
- **Surprised** has the highest per-class accuracy (50%) but is also the most common false prediction.
- High-arousal emotions (Angry, Happy) are mutually confusable, as are low-arousal emotions (Calm, Sad, Neutral).

These patterns are expected when training on only 2 actors. With 19 training actors, the model will learn more robust emotion-specific features and reduce these confusions.

F. Implementation Details

IMPLEMENTATION DETAILS	
Component	Detail
Language	Python 3.12.13
Audio loading	librosa 0.11.0
Machine learning	scikit-learn 1.8.0
Data handling	NumPy, Pandas
Visualization	Matplotlib, Seaborn
Model persistence	joblib
Environment	Conda on WSL2 (Ubuntu)
Hardware	4 CPU cores, 8.3 GB RAM
Pipeline	t-tracked Jupyter Notebooks

6. PLANNED WORK AND TIMELINE

A. Remaining Experiments

1) Phase A: Feature Engineering (Notebook 04):

1) Log-Mel Spectrograms for CNN

- Input: trimmed WAVs from data/{train,val,test}/{speech,song}/
- Compute log-mel spectrogram (n mels=128, n fft=1024, hop length=512)
- Output shape: (128, 94)→ save as .npy arrays
- Separate files for speech (1,440), song (1,012), combined (2,452)

2) MFCC Vectors for SVM

- Extract 40 MFCCs
- Aggregate as mean, std, min, max→40*4 = 160-dim

3) Save Label Arrays

- Emotion class ints for each clip
- One-hot optional for CNN training

2) Phase B: Model Training (Notebook 05):

4) Train SVM Baseline

- Speech-only 8-class: 1,140 files
- Song-only 6-class: 792 files
- Combined 8-class: 1,932 files
- Grid search $C=[0.1,1,10,100]$

5) Train 1D CNN

- 3-block PyTorch architecture • Adam (lr=10⁻³), batch=32
- Early stopping (patience=10)

6) Cross-Speaker Analysis

- Per-speaker accuracy (Actors 23–24)
- Gender+intensity breakdown

3) Phase C: Cross-Channel and Ablation (Notebook 06):

7) Cross-Channel Experiments

- Train on speech, test on song
- Train on song, test on speech
- Report accuracy degradation

8) Ablation Experiments

- Vary n_mels: [64, 128, 256]
- Vary conv blocks: [2, 3, 4]
- With/without BatchNorm, Dropout

4) Phase D: Deployment (Notebook 07):

9) Deployment Evaluation

- CPU inference latency+model size
- Export as TorchScript

10) Final Report and Presentation

- IEEE-format final report
- Presentation slides

B. Risks and Mitigation

TABLE XI
RISKS AND MITIGATION

Risk	Like.	Imp.	Mitigation
Overfitting	High	High	Dropout, weight decay, early stopping
Poor- cross speaker	Med.	High	Speaker-disjoint splits;

			per-speaker results
CNN<SVM	Low	High	Wider CNN (512 ch); 4th conv block
Class imbalance	Med.	Med.	Weighted loss; per-class F1
Song<Speech Med.		Med.	Expected; cross channel inherently harder

C. Timeline

PROJECT TIMELINE			
Phase	Tasks	Deliverables	
Midterm (done)	Data, preprocessing, midterm report	EDA, split_labels.csv, this report	data,
Phase A	Feature extraction	.npy arrays for 2,452 clips	
Phase B	SVM+CNN training	Models, curves, metrics	
Phase C	Cross-channel, ablation	Accuracy table, ablation results	
Phase D	Deployment, final report	Model export, slides	PDF,

7. TEAM CONTRIBUTIONS

A. Elsayed—Audio Preprocessing and Baseline

1) Completed:

- Downloaded and verified RAVDESS dataset (1440 audio files).
- Built filename parser to extract emotion, intensity, statement, repetition, actor, and gender labels.
- Implemented preprocessing pipeline: silence trimming, amplitude normalization, fixed-length padding.
- Ran exploratory data analysis: class distribution, clip durations, waveform and spectrogram visualization.

2) Planned:

- Finalize MFCC feature extraction pipeline.
- Implement and train SVM baseline with grid search.
- Implement data augmentation (time shift, noise injection, speed perturbation).

B. Salma—CNN Implementation and Training

1) Completed:

- Designed 1D CNN architecture (3 conv blocks, global average pooling, fully connected head).
- Implemented log-mel spectrogram extraction prototype.
- Verified feature shapes and caching pipeline.

2) Planned:

- Implement full CNN training loop in PyTorch (loss, optimizer, scheduler, early stopping).
- Run training experiments with hyperparameter tuning.
- Run ablation experiments (number of mel bands, conv blocks, regularization).

C. Khaled—Evaluation and Report

1) Completed:

- Defined evaluation metrics: accuracy, per-class F1, confusion matrix, latency, model size.
- Set up speaker-disjoint train/validation/test split (Actors 1–18/19–21/22–24).
- Drafted midterm report.

2) Planned:

- Run full evaluation on test set for both CNN and SVM.
- Cross-speaker analysis: per-speaker accuracy, gender breakdown, intensity comparison.
- Measure inference latency and model size for deployment report.
- Compile final report and presentation.

8. REFERENCES

- [1] S. R. Livingstone and F. A. Russo, “The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS),” *PLOS ONE*, vol. 13, no. 5, p. e0196391, 2018.
- [2] S. R. Livingstone and F. A. Russo, “RAVDESS Dataset,” Zenodo, 2018. doi: 10.5281/zenodo.1188976
- [3] M. Neumann and N. Q. K. Vu, “Attentive Convolutional Neural Network based Speech Emotion Recognition,” *arXiv preprint arXiv:1706.00612*, 2017.
- [4] S. Tripathi *et al.*, “Deep Neural Networks for Emotion Recognition in Speech,” in *Proc. IJCNN*, 2019.

9. Use of GenAI Tools

GenAI tools (Claude / ChatGPT) were used in limited capacities for this project. Specifically, they were used to (1) correct grammar and phrasing in the written report (2) Clarify the documentation of the Python libraries used (librosa, scikit-learn), (3) Assist with debugging after initial self-directed attempts. No end-to-end code was generated through GenAI and no report sections, experimental results, or figures were generated by generative AI. All model design choices, experimental choices, and written analysis are the team's own.